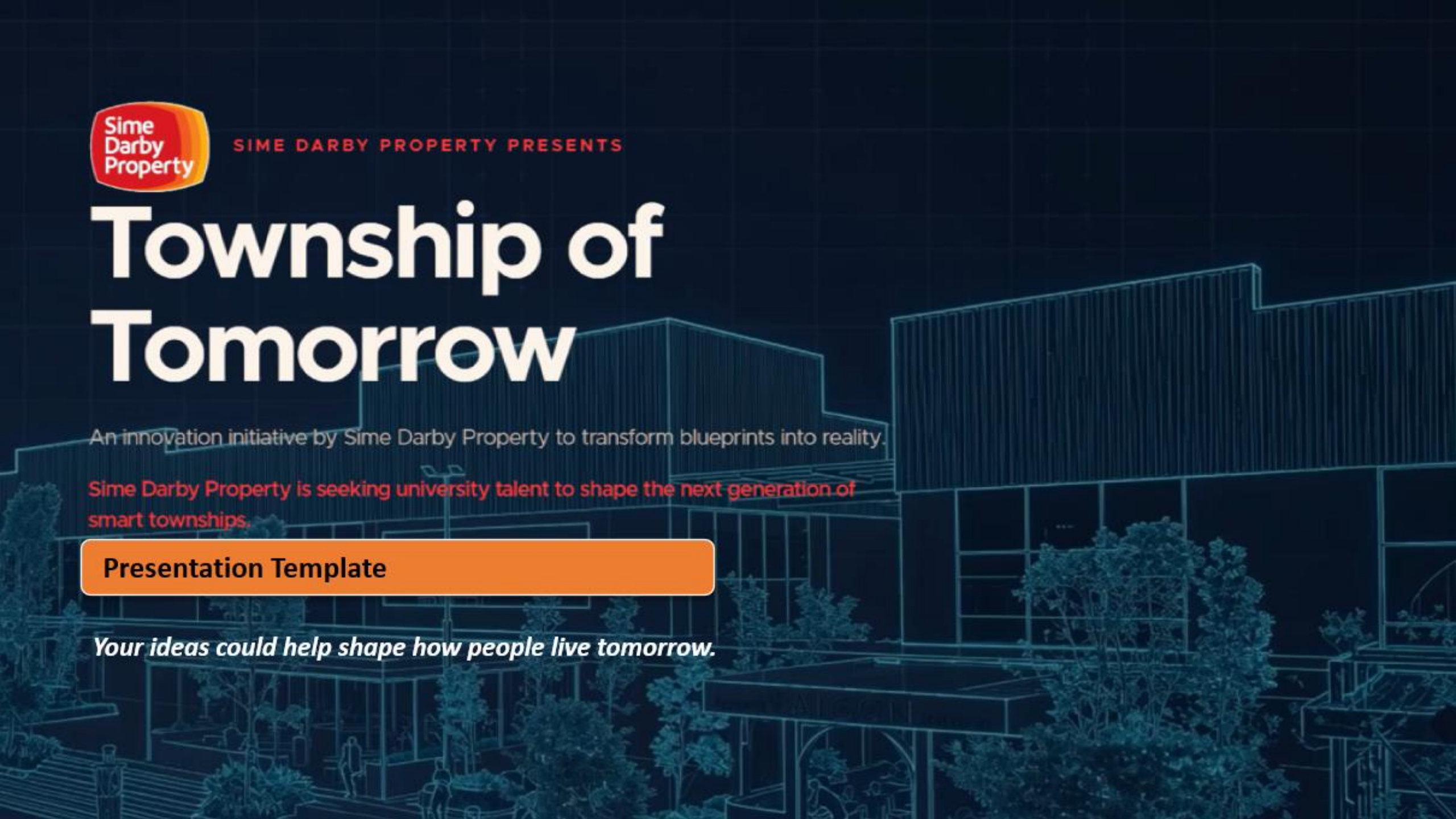

Sime Darby Property
SIME DARBY PROPERTY PRESENTS
Township of Tomorrow
An innovation initiative by Sime Darby Property to transform blueprints into reality.
Sime Darby Property is seeking university talent to shape the next generation of smart townships.
Presentation Template
Your ideas could help shape how people live tomorrow.

GreenLoop

Smart Recycling Ecosystem · recycling that pays for itself

Team Information

- **Team Name: Synkosys**
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  - Leeya Sofea binti Mohd Rizal Izwan, Universiti Teknologi MARA · leeyasofea199@gmail.com
- **Live demo: greenloop-tan.vercel.app · Code: github.com/NajibAzir/greenloop**

1. The Problem

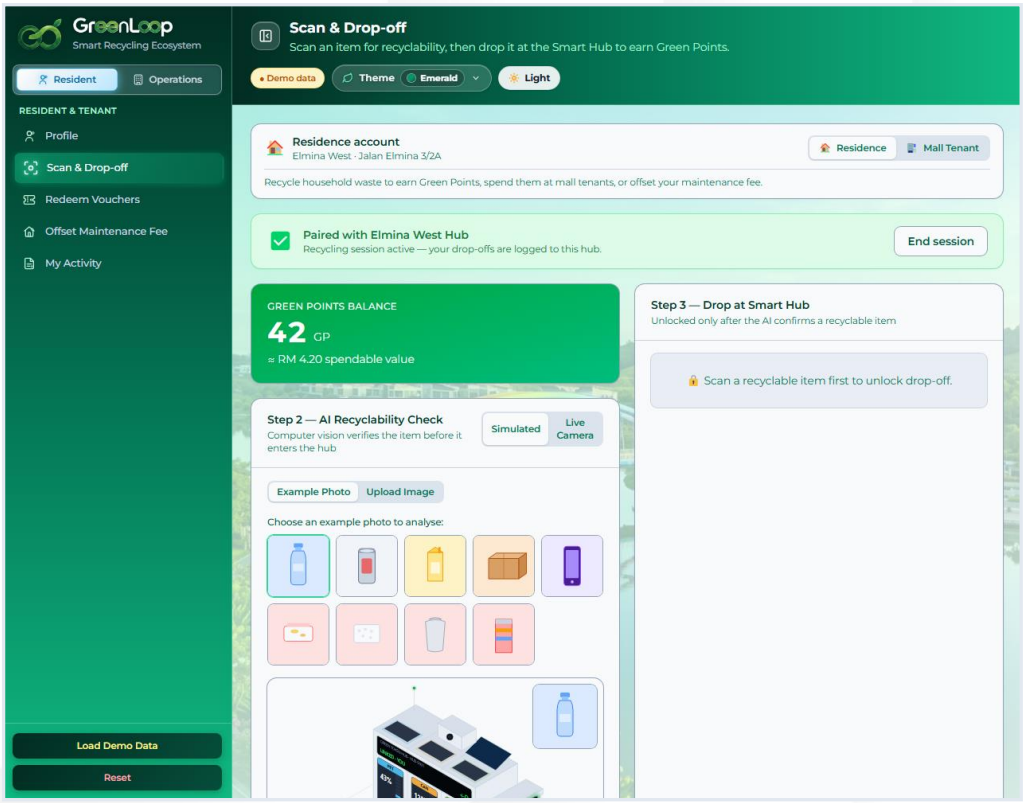
- **Malaysia wants to recycle more, but most recyclable waste still ends up in landfill.**
  - ~38,000 tonnes of solid waste are generated every day; about 82% still goes to landfill.
  - The national recycling rate sits around 30%, below the 40%-by-2025 target, yet ~40% of landfilled waste is recyclable.
- **In a township it stalls for three reasons:**
  - Residents have no real incentive, effort, no reward, so turnout stays low.
  - Recycling is treated as a cost centre, so programmes get cut.
  - Collection is blind, trucks run fixed routes to half-empty bins, wasting fuel.
- Sources: UNIDO (2025); Asia News Network (2026); KPKT Circular Economy Blueprint 2025 to 2035. Figures indicative.

2. Target User

- **Three groups, one loop, modelled on real Sime Darby Property communities.**
  - Residents (households), want convenient recycling and an everyday reward.
  - Mall tenants (F&B / retail, e.g. Elmina Lakeside Mall), want to offset monthly service charges from packaging waste.
  - Developer operations (Sime Darby Property), want visibility, lower haulage cost and measurable ESG outcomes.
- Grounded in real locations: City of Elmina (Elmina West / East, Denai Alam, Bukit Subang), Elmina Lakeside Mall, Bandar Bukit Raja.

3. Your Solution

- **GreenLoop closes the loop: recycling becomes a self-funding economy.**
  - Scan & earn, AI verifies the item; residents earn Green Points (GP).
  - Spend like cash, redeem GP at mall tenants or offset monthly fees.
  - Funded by savings, avoided haulage + material sales + carbon value pay for it.
  - Collect smart, AI dispatches trucks only to full bins.



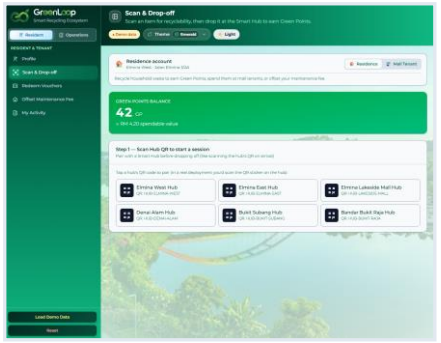
Resident Scan & Drop-off (live prototype)

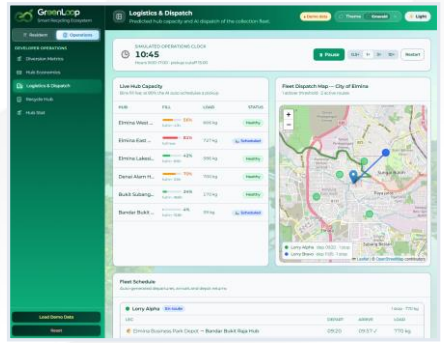
4. Innovation

- **A funded, verified, intelligent loop, not just another points app.**
  - Rewards are funded by avoided haulage + material sales + carbon value (not a subsidy).
  - AI vision verifies each item before it is accepted (vs “trust me, it's recyclable”).
  - Live AI dispatch sends trucks only to full bins, on real road routing (vs fixed routes).
  - Residents and mall tenants are served by one engine (vs a single user type).

5. How it Works

- **Residents → Hubs → Operators → Rewards, one closed loop:**
  - 1. Scan, resident scans an item; AI vision confirms it's recyclable.
  - 2. Drop at Smart Hub, pair via hub QR; item accepted, GP credited instantly.
  - 3. AI Dispatch, bin fills; at 80% a truck is scheduled to full bins only.
  - 4. Recover & Sell, material is diverted from landfill and sold to recyclers.
  - 5. Value funds the loop, savings + sales + carbon fund the GP and fee offsets, and the loop repeats.



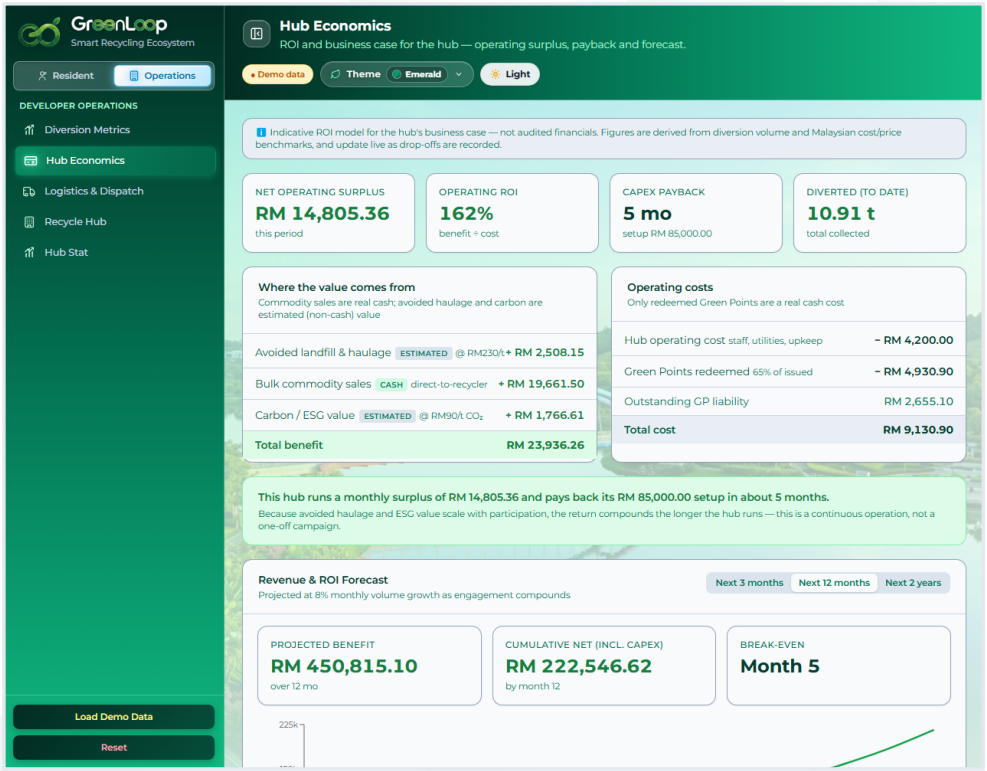


6. Impact

- **Measurable outcomes, honest about what's demonstrated vs projected.**
  - Environmental, tonnes diverted from landfill; CO_2e avoided; supports the 40% target.
  - Community, residents earn GP; GP offsets fees; higher, sustained participation.
  - Operational, lower haulage cost, live hub/fleet visibility, a self-funding programme.
- Demonstrated: working AI scan, GP economy, live dispatch, 6-hub map. Modelled: ~10.9 t/month diverted and an operating surplus. Needs validation: bin hardware, real haulage cost, recycler & carbon prices.

7. Feasibility

- **The hub pays for itself, transparent monthly economics (indicative, reconciles with the live dashboard).**
  - Benefit: material sales RM 19,662 (cash) + avoided haulage RM 2,508 + carbon RM 1,767 = RM 23,936.
  - Cost: operating RM 4,200 + Green Points redeemed RM 4,931 = RM 9,131.
  - **Net cash flow ≈ RM 14,805 / month → RM 85,000 setup ÷ 14,805 ≈ 5.7-month simple payback.**
  - Conservative case (no carbon): net ≈ RM 13,000/mo, payback ~6.5 months. Still self-funding.
- RM230/t avoided-haulage is an assumption pending validation with Sime Darby's waste contractor.

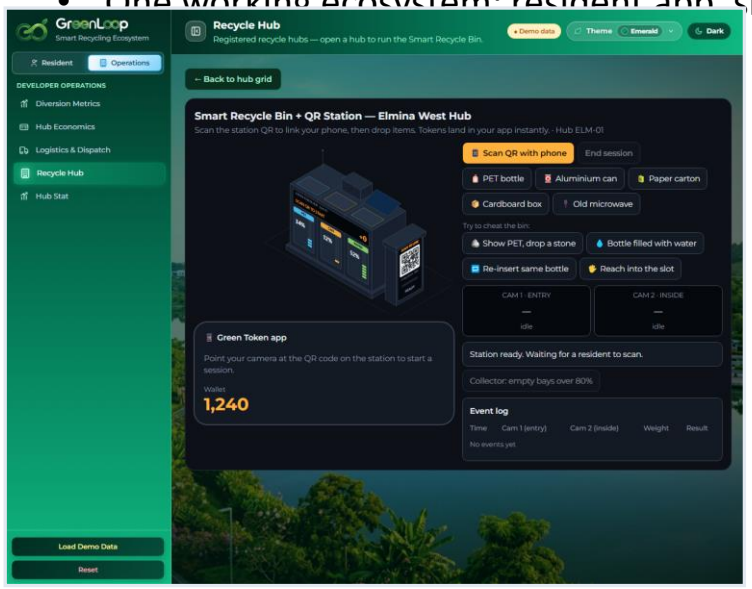


7. Implementation

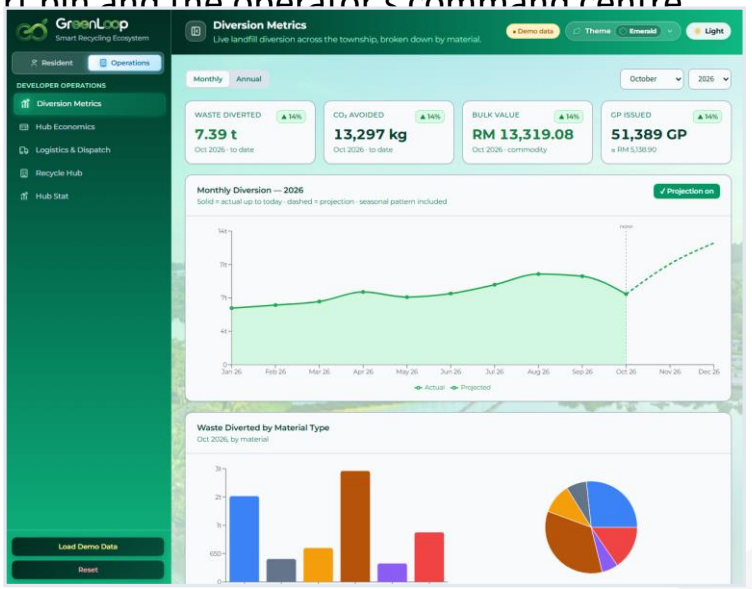
- **Start with one hub at Elmina West: prove it, then scale precinct by precinct.**
  - Month 1, install & integrate: Smart Hub + bin hardware; link rewards and resident sign-up.
  - Month 2-3, run & measure: track diversion, participation and real cost vs the model.
  - Month 4, review (go / no-go): compare actuals to assumptions; confirm payback.
  - Month 5+, scale: roll out to Elmina East, Denai Alam, Elmina Lakeside Mall.
- Stakeholders: Sime Darby Property ops, waste contractor, mall tenants, residents, recyclers. Validate first: bin hardware, real RM/t haulage, recycler prices, POS integration, resident identity & PDPA.

7. Visual Concept

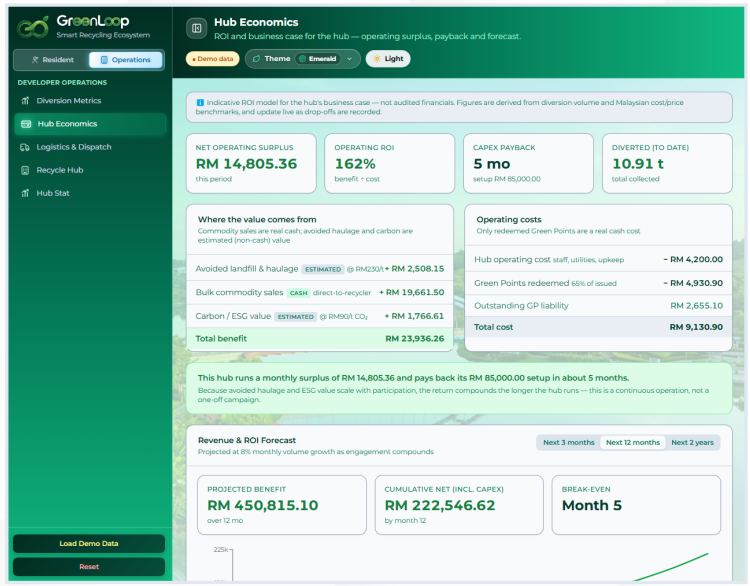
- One working ecosystem: resident app, smart bin and the operator's command centre



Smart Recycle Bin + QR station



Diversion metrics (actual vs projected)



Hub economics (business case)

8. Why Your Team Should Win

- GreenLoop connects AI-assisted recycling, resident rewards and smarter collection into one township ecosystem, tackling both participation and the cost of collection, not just one of them.
- We built a deployed, interactive prototype and a one-hub pilot model for Elmina West, and we are transparent about what is demonstrated versus what still needs real-world validation.
- **Our ask: a one-hub pilot at Elmina West plus access to operational data, so we can measure actual diversion, collection savings and payback before scaling.**

Thank You